



GitHub Copilot Dev Days

CLI *FTW!*



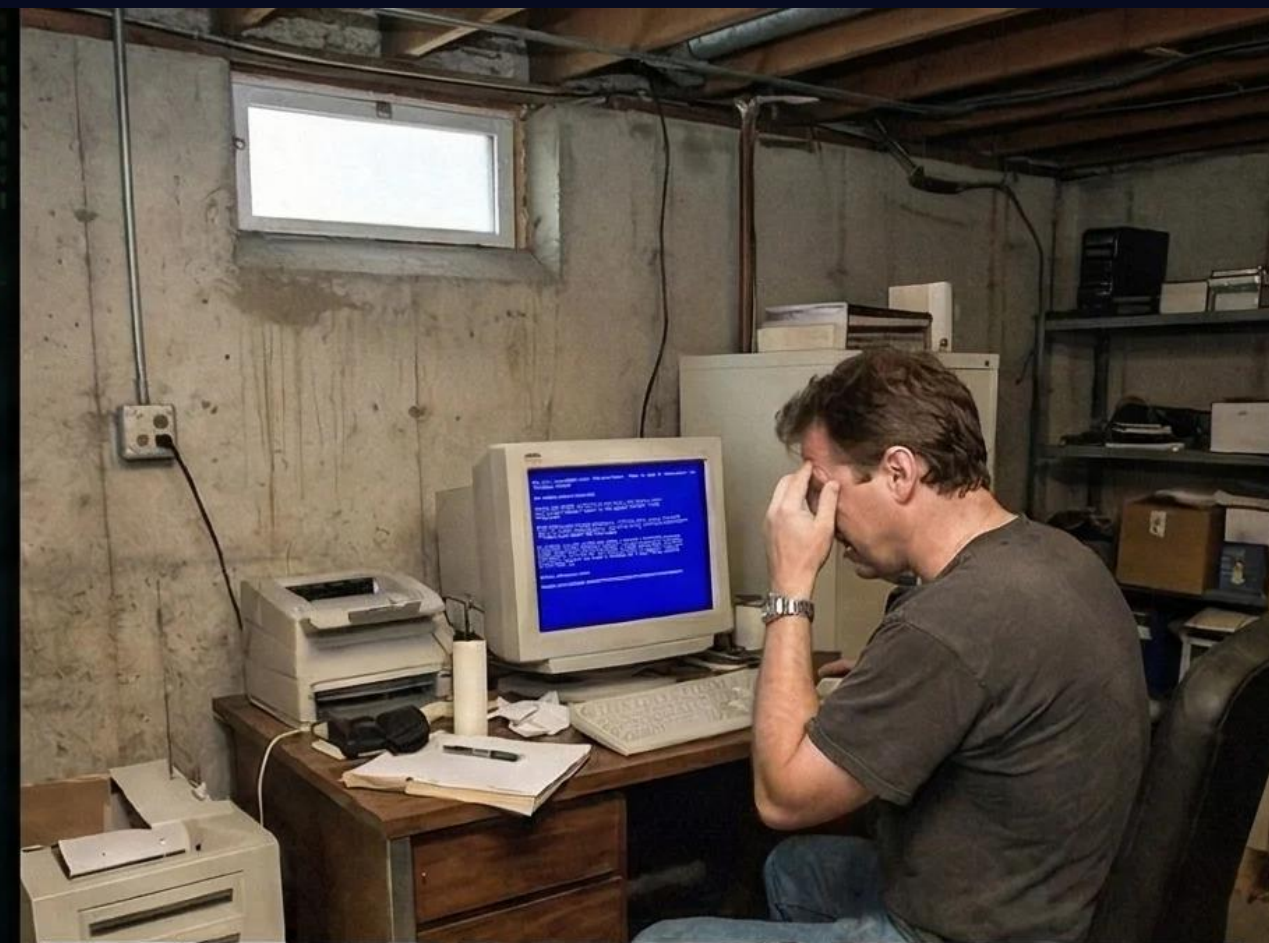
Jim McKeeth

BSDG.dev - BoiseCodeCamp.com

jim@mckeeth.org



WHAT MY FRIENDS THINK I DO



WHAT MY PARENTS THINK I DO



WHAT SOCIETY THINKS I DO



WHAT I THINK I DO



WHAT MY BOSS THINKS I DO



WHAT I REALLY DO

Confirm Delete

 Confirm Delete

data streams =206:322.575.121:30
data visuation = 325.385.93.26799325
stouten-connectors:13
data-vtualized:15r:0.090.15
data-menten[020]:92863
data-onnected:iden

Terminal

\$ ls -la
> total 48
d drwx-x-x 2 staff 4096 2:36 .
d drwx-x-x 2 staff 4096 2:06 ..
d drwx-x-x 3
> drwxr-x 3
> drwxr-x 3
> drwxr-x 2
> drwxr-x 3
> drwxr-x 2
\$

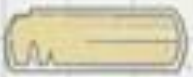
Terminal

\$ ls -la
> cd /home/user
> git commit -m 'Initial commit'
> git none
> git commit -m 'Initial commit'
\$

Terminal

\$ ps aux
> cd h

streamer



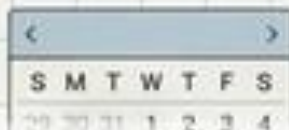
Confirm Delete

```
function pseudo(oesnd)  
  for pseudeconnst:  
    for ( = 0 0) {  
      cons := emit('nitial_c  
      if (is != null {  
        const.print("Patched  
        node +> 11;  
        oode<<< "%&set npr  
      }  
      return node;  
    }  
  }  
  pseudo-code;
```

stop bothering me,
or I shall replace
you with a very small
shell script.

New Project Wizard

Template:
- New Project
- Project Template
- Momproje
- Project Project





```
if [ -f "$file" ]; then
  echo "Processing $file";
fi
#!/bin/bash
while read line; do
  grep "pattern" "$line"
done
variable=$(command)
ls -l | awk
{{print $9}}
```

```
#!/bin/bash
if [ -f "$file" ]; then
  echo "Processing $file";
fi
variable=$(command)
ls -l | awk
{{print $9}}
```

```
#!/bin/bash
if [ -f "$file" ]; then
  echo "Processing $file";
fi
variable=$(command)
ls -l | awk
{{print $9}}
```

```
#!/bin/bash
if [ -f "$file" ]; then
  echo "Processing $file";
fi
variable=$(command)
ls -l | awk
{{print $9}}
```

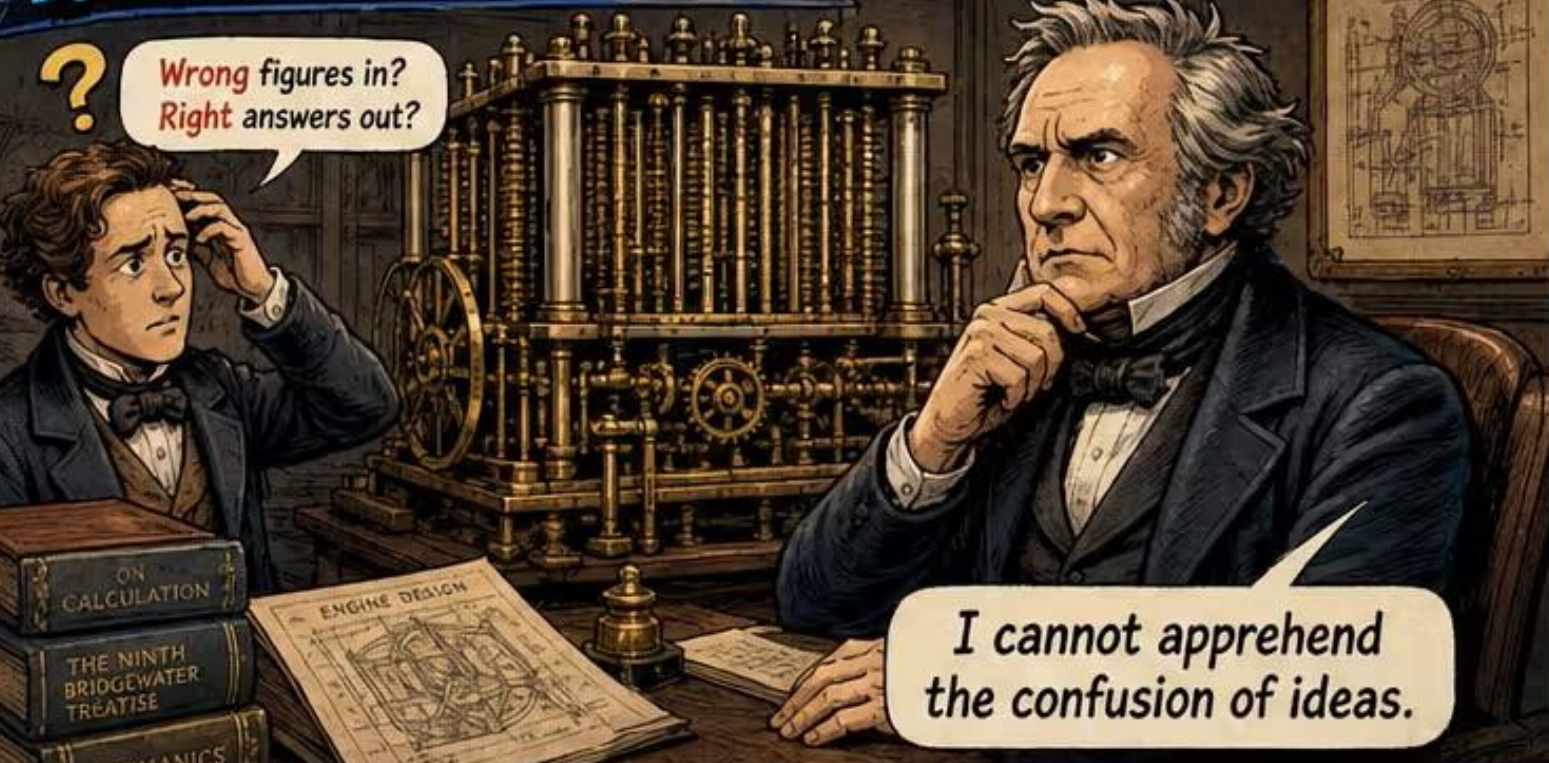
Thanks to Vibe Coding with an AI Coding Agent

English is the new
PROGRAMMING LANGUAGE

Except English is imprecise and generally a mess.

1. BABBAGE SAW IT FIRST

? Wrong figures in?
Right answers out?



I cannot apprehend the confusion of ideas.

2. NOVICE PROMPTS LOSE SIGNAL

Missing vocabulary and structure hurts code generation.

NOVICE PROMPT

make app better fix stuff do magic

OUTPUT (NOVICE) ⚠

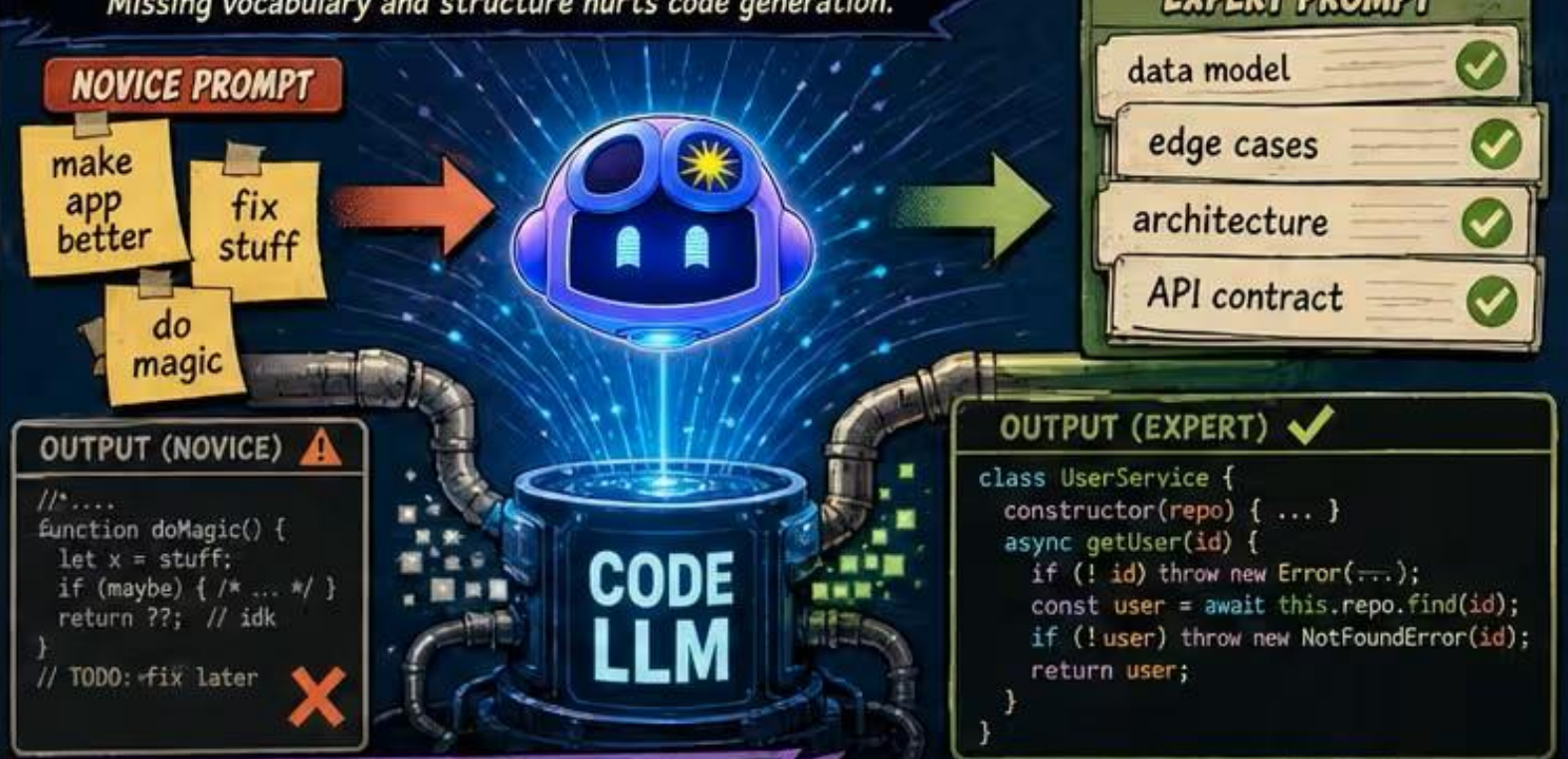
```
/*....  
function doMagic() {  
  let x = stuff;  
  if (maybe) { /* ... */ }  
  return ??; // idk  
}  
// TODO: fix later
```

EXPERT PROMPT

- data model ✓
- edge cases ✓
- architecture ✓
- API contract ✓

OUTPUT (EXPERT) ✓

```
class UserService {  
  constructor(repo) { ... }  
  async getUser(id) {  
    if (!id) throw new Error(...);  
    const user = await this.repo.find(id);  
    if (!user) throw new NotFoundError(id);  
    return user;  
  }  
}
```



3. REQUIREMENTS BEAT TRICKS

Input quality tracks output quality.

PROMPT TRICKS

- make it amazing ✨
- use best practices!! ✨
- be creative 😊
- optimize it 🧠
- think step by step
- magic words here...

vibes hype fluff

OUTPUT (TRICKS) ⚠

```
// generic solution  
// missing validation  
// edge cases ignored  
// not maintainable
```

CLEAR REQUIREMENTS

- Goal: User signup flow
- Constraints: auth, rate limit
- Data model: User, Session
- Edge cases: duplicates, lockout
- Acceptance tests:
 - valid signup succeeds
 - duplicate email rejected
 - rate limit enforced

OUTPUT (REQUIREMENTS) ✓

```
async function signup(dto) {  
  validate(dto);  
  if (await userExists(dto.email))  
    throw new ConflictError('email');  
  enforceRateLimit(dto, 10);  
  const user = await createUser(dto);  
  await createSession(user.id);  
}
```



4. DOMAIN KNOWLEDGE GROUNDS THE MODEL

Specific context improves technical accuracy.

DOMAIN KNOWLEDGE

- TERMINOLOGY**
 - event sourcing
 - aggregate root
 - idempotency
- FORMULAS**
$$PV = \sum_{t=1}^n \frac{CF_t}{(1+r)^t}$$
- CONSTRAINTS**
 - ACID compliance
 - latency < 200ms
 - PII encryption
- EXAMPLES**
 - OrderCreated event
 - PaymentFailed event

ENRICHED PROMPT

Design an event-driven order processing service.

Use event sourcing. Ensure idempotent handlers. Handle out-of-order events.

Provide models, key flows, and pseudo-code.

ACCURATE OUTPUT ✓

```
// Idempotent handler  
if (!processed(event.id)) return;  
  
// Apply domain rules  
const order = loadAgg(event.aggregateId);  
order.apply(event);  
markProcessed(event.id);
```

HALLUCINATIONS REDUCED

use Blockchain! ✗ just call 3rd party API! ✗ store in CSV file! ✗

POP!



BETTER CONTEXT BEATS BETTER VIBES.



[w.wiki/55\\$A](https://www.wiki/55$A)



doi.org/10.18653/v1/2024.findings-acl.501



doi.org/10.1145/3731756



doi.org/10.2139/ssrn.4790854

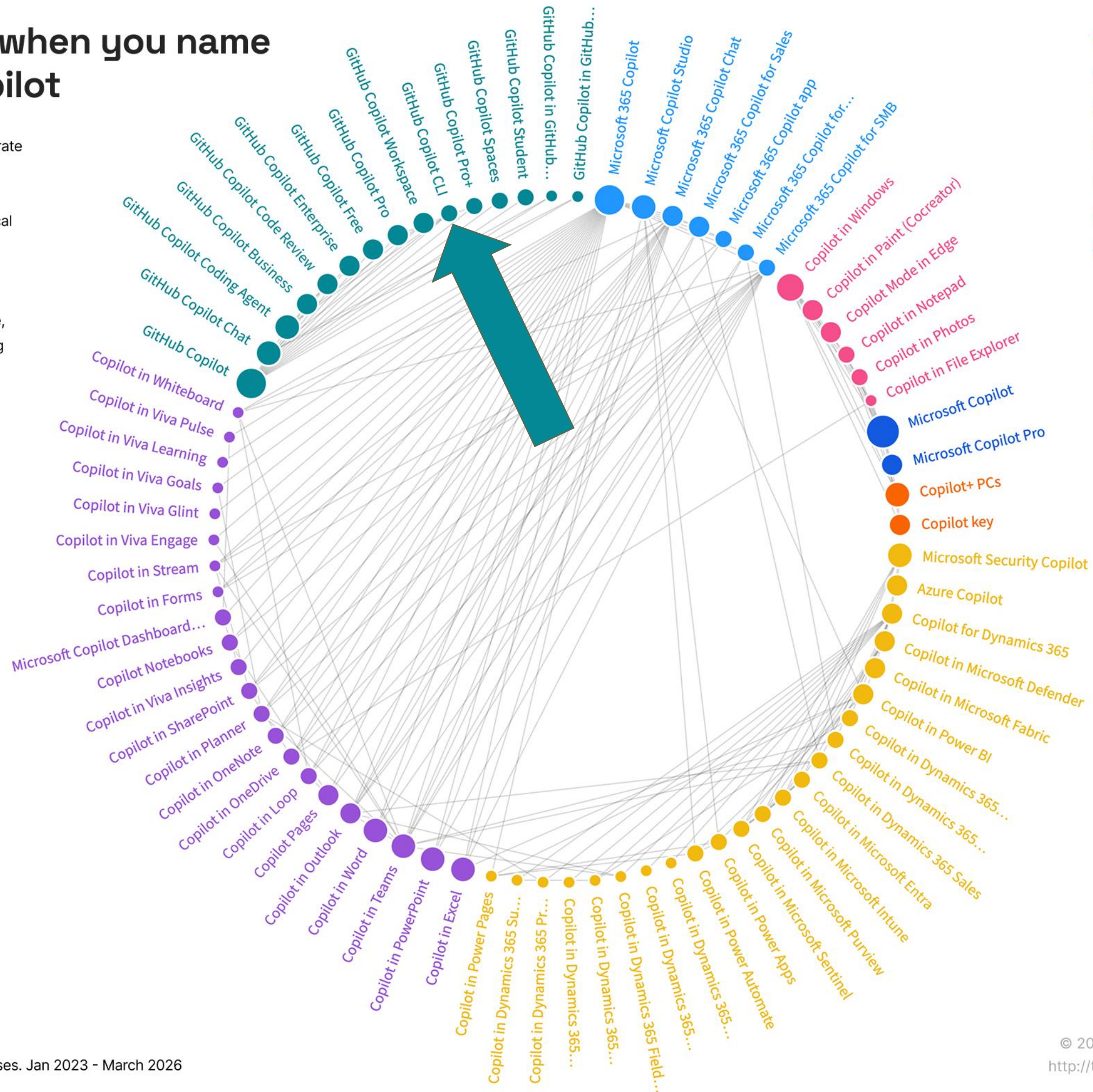
What happens when you name everything Copilot

As of April 2026, Microsoft has 78 separate products and tools all named “Copilot”. There are now Copilots inside Copilots, Copilots for other Copilots, and a physical Copilot key on your keyboard for summoning them.

This visualisation maps every single one, grouped by category, with lines showing Copilots that overlap, bundle together, or sit inside one another.

Dot size approximates relative prominence based on Google search volumes and press coverage.

Check out the interactive version at
<http://teybannerman.com>





<> What's in the box

Core Features

- Interactive Assistance
- Headless Automation
- Code Generation & Modification
- Tool Execution & Shell Commands
- Multi-file context awareness
- Lifecycle event hooks
- Modes for multiple situations
- MCP server integration

```
skills — GitHub Copilot — copilot < node /opt/homebrew/bin/copilot --yolo — 119x36

[patrick@FC7Q0F77YL skills % copilot --yolo ]

GitHub Copilot v0.0.400
Describe a task to get started.

Pick a model with /model. Delegate changes with an AI-generated PR using /delegate.
Enter ? to see all commands. Copilot uses AI, so always check for mistakes.

• 💡 No copilot instructions found. Run /init to generate a copilot-instructions.md file for this project.

~/projects/skills[/ feature*]                               claude-opus-4.5 (3x)

> /add-dir

  /add-dir <directory>      Add a directory to the allowed list for file access
  /agent                    Browse and select from available agents (if any)
  /allow-all, /yolo        Enable all permissions (tools, paths, and URLs)
  /clear, /new              Clear the conversation history
  /compact                  Summarize conversation history to reduce context window usage
  /context                  Show context window token usage and visualization
  /cwd, /cd [directory]     Change working directory or show current directory
  /delegate [prompt]        Delegate changes to remote repository with AI-generated PR
  /diff                     Review the changes made in the current directory
  /exit, /quit              Exit the CLI
```

GitHub Copilot CLI

Installation



macOS / Linux (Homebrew)
`brew install copilot-cli`



Windows (WinGet)
`winget install GitHub.Copilot`



Any OS (npm, Node.js ≥ 22)
`npm install -g @github/copilot`



Many modes. **Many use cases.**



`copilot`

User runs CLI locally on their machine with full interactive UI and prompts

Interactive



`copilot -p`

Non-interactive execution for automation pipelines and CI/CD workflows

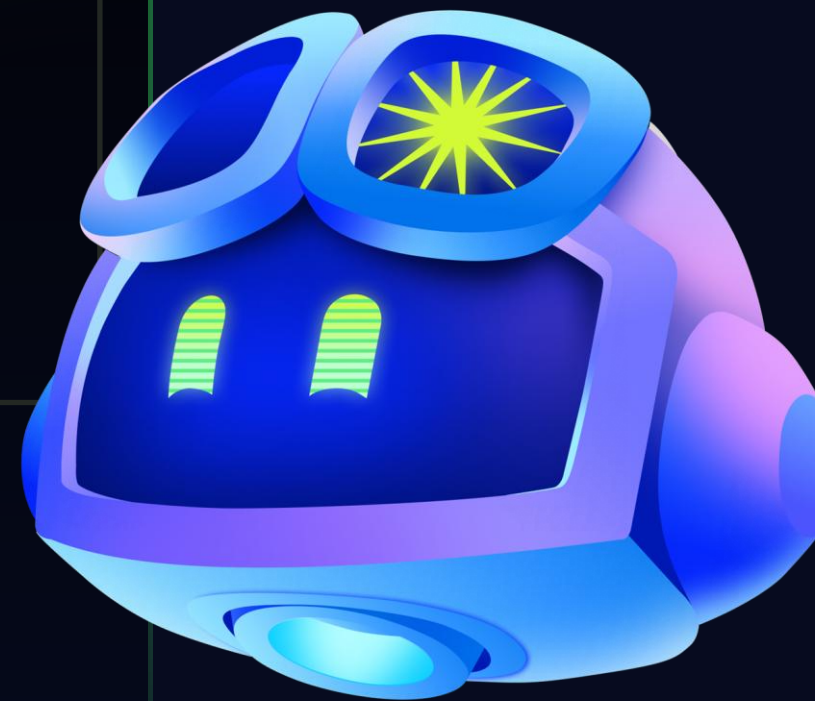
Headless



`copilot --server`

Exposes JSON-RPC port for external clients to drive CLI functionality remotely via the Copilot SDK

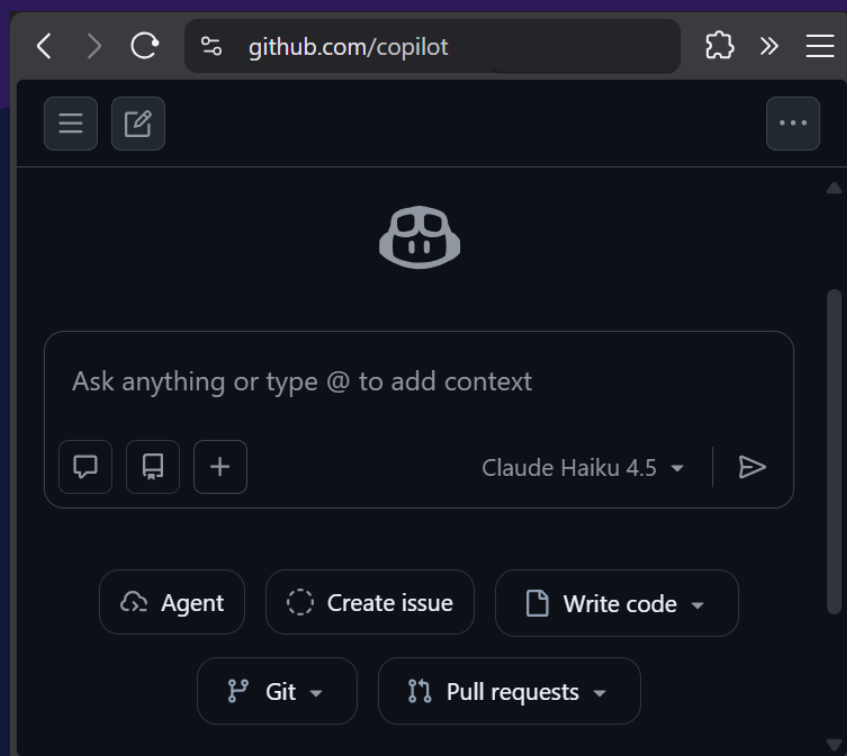
Server





Effort

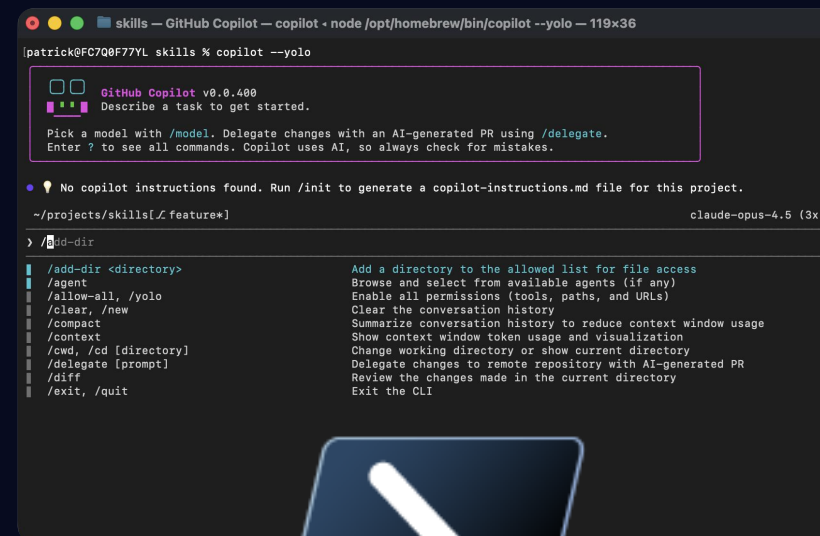
Power & Flexibility



Web



IDE



CLI



SDK, etc.

Which LLM To use?

- They are very similar.
- Frequent updates changes *ranking*.
- Skill.md and Agents.md are standards.
- *Use what works for you.*

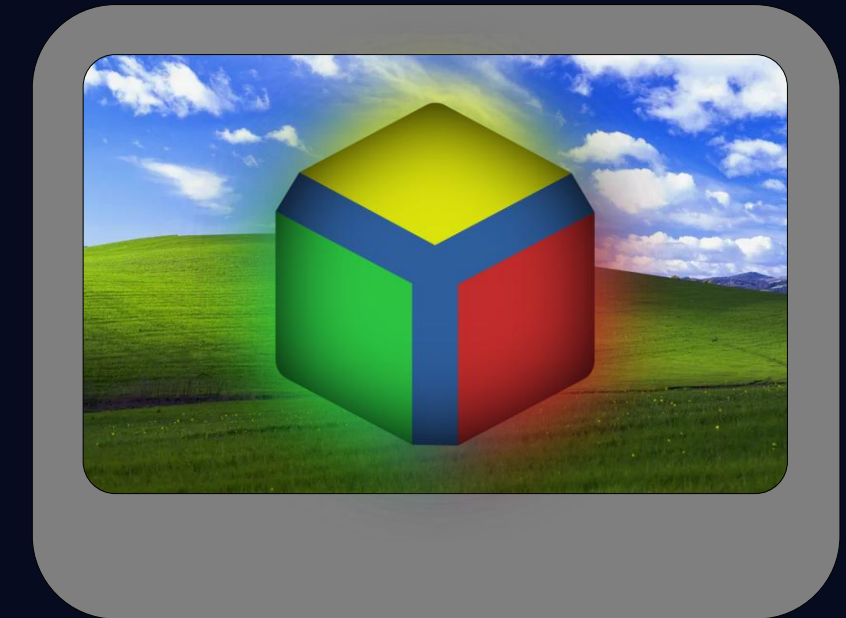




Configure a Database



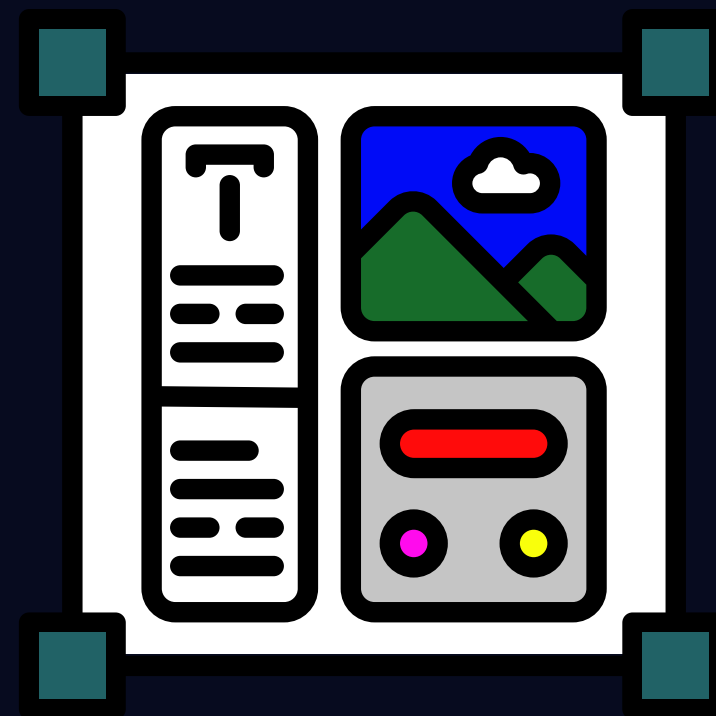
SSH to remote machine



Start, Stop, Create, Revert a VM



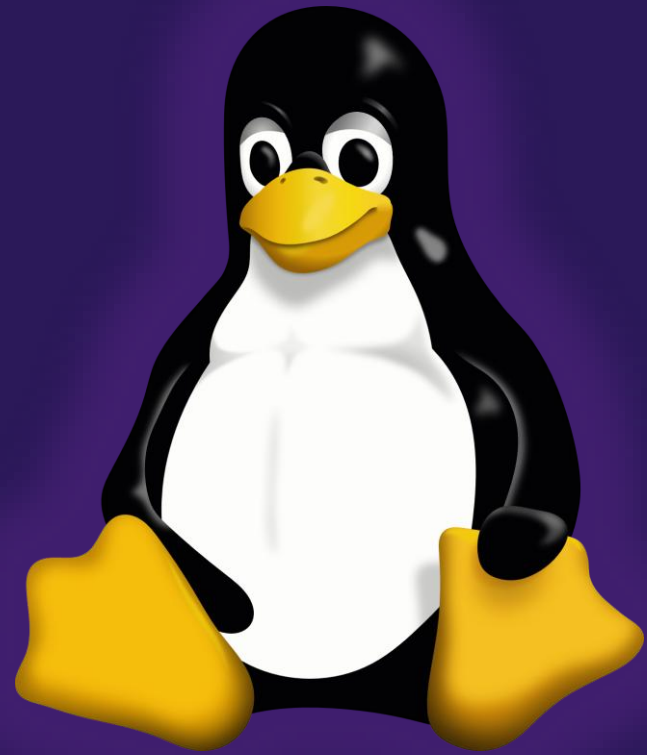
Run a Program



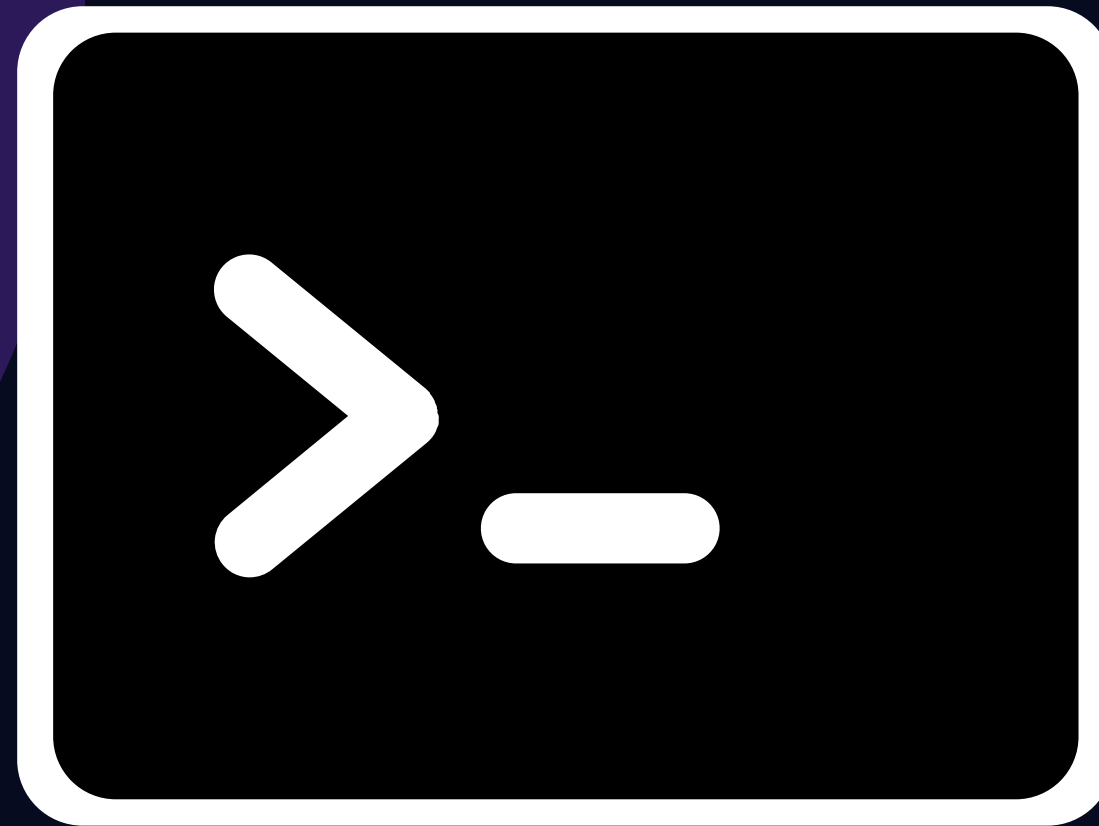
Layout a Document



Start or Stop A service



Linux via WSL



Anything you can do
From the Terminal

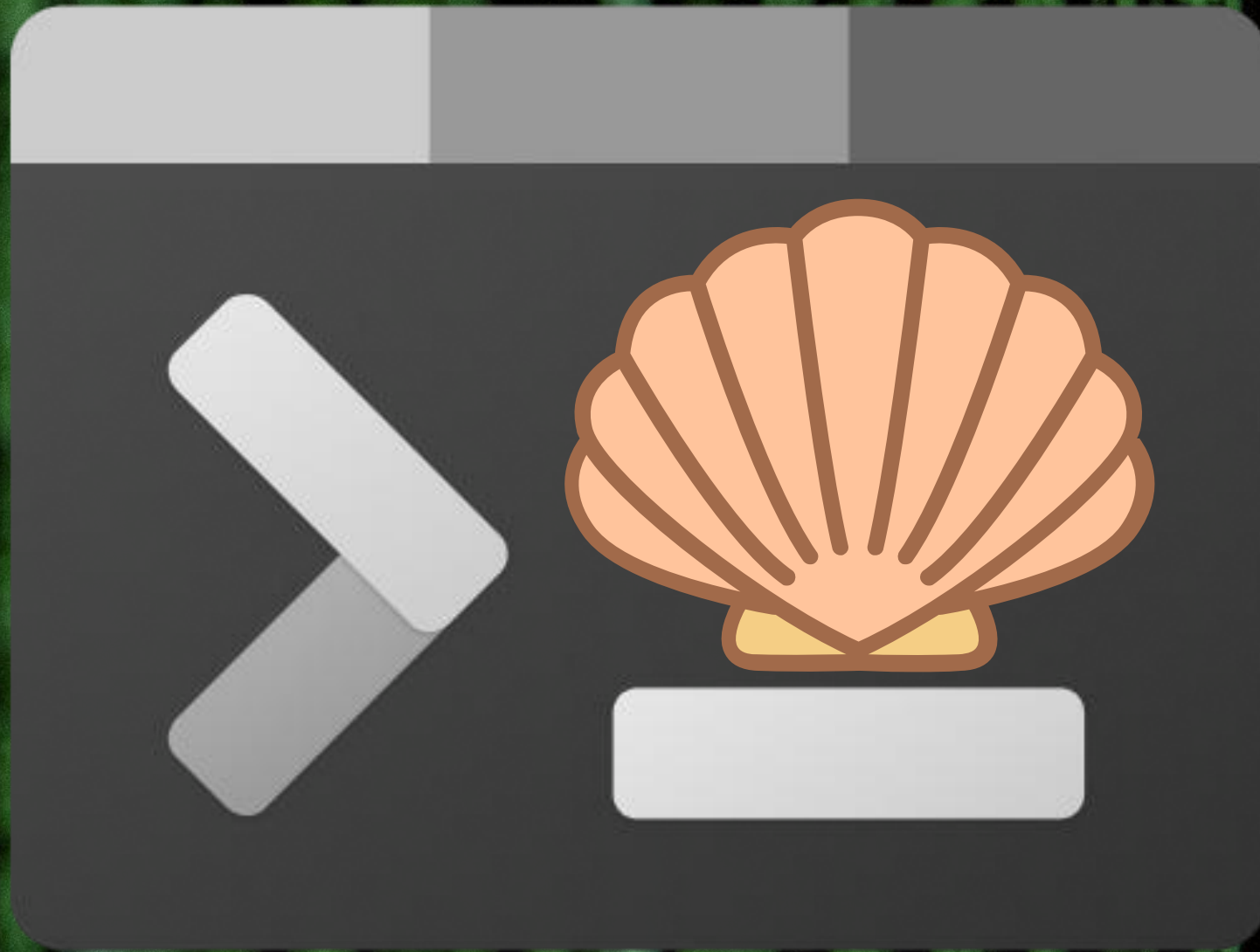


Powershell



Git Bash





A Shell runs *in* a Terminal

CLI: Command Line Interface

TUI: Terminal User Interface

REPL: Read–eval–print loop

- Terminals
 - Microsoft's github.com/microsoft/terminal
 - Cross-platform wezterm.org
- PowerShell 7
 - github.com/PowerShell/PowerShell
 - learn.microsoft.com/powershell/scripting
- Git Bash gitforwindows.org
- TUI Editors
 - github.com/microsoft/edit
 - github.com/sharkdp/bat
- PowerShell Upgrades
 - [OhMyPosh.dev](https://ohmyposh.dev)
 - [NerdFonts.com](https://nerd-fonts.com)
 - github.com/PSModule/NerdFonts

Fresh

getFresh.dev

A powerful terminal text editor and IDE.

Intuitive keys & mouse · Fast & light · Huge files · Themes · TypeScript plugins · Open source
· [and more!](#)

Linux

macOS

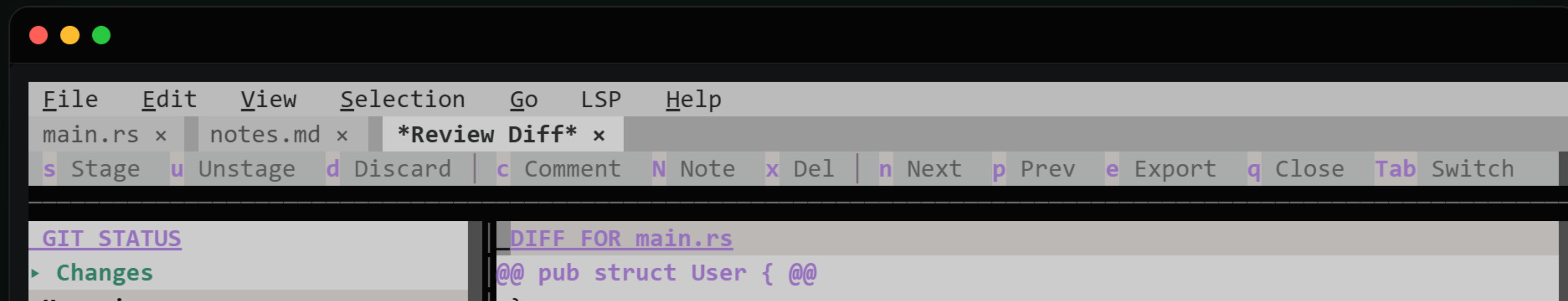
Windows

Install:

```
winget install fresh-editor
```



[More options...](#)





<https://github.com/sharkdp/bat>

► bat test.md

```
File test.md
1  # Markdown example
2
3  Note how we can correctly syntax-highlight the
4  code blocks *inside* the Markdown document.
5
6  Python example:
7
8  ``` python
9  cmd = "bat"
10 print("Hello from {}".format(cmd))
11 ```
12
13 Ruby example:
14
15 ``` ruby
16 cmd = "bat"
17 puts "Hello from #{cmd}"
18 ```
```

► bat -A non-printable.txt

```
File: non-printable.txt
1  .....space-indented•line LF
2  | | |tab-indented•line LF
3  | .....mixed-indented•line LF
4  LF
5  backspace•character•in•quotes:•"BS" LF
6  escape•character•in•quotes:•"ESC" LF
7  LF
8  line•with•trailing•space••• LF
9  LF
10 line•that•ends•with•carriage•return CRLF
11 last•line LF
```

► bat src/index.html

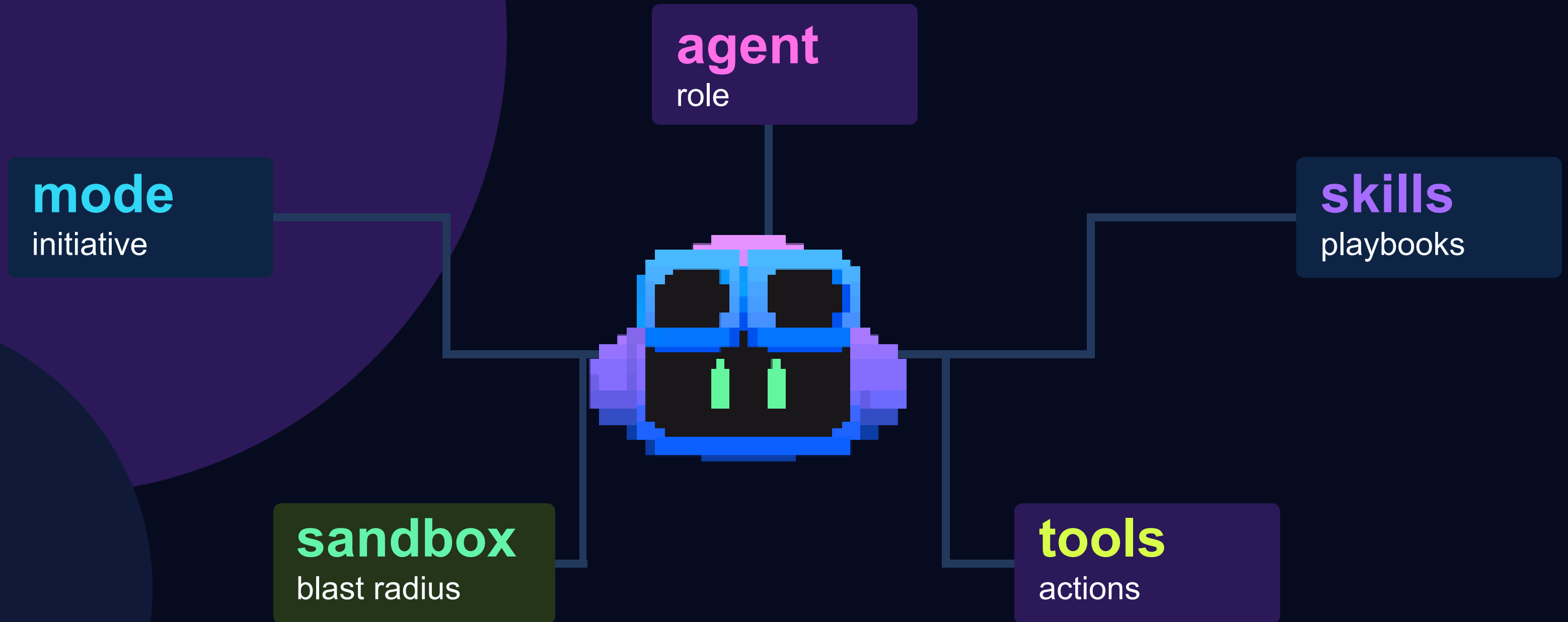
```
File src/index.html
1  <!DOCTYPE html>
2  <html lang="en">
3      <head>
4          <meta charset="utf-8">
5      ~      <title>a changed title</title>
6          <link rel="stylesheet" href="style.css">
7      </head>
8      <body>
9  +          New lines that have been
10 +          added since last commit.
11      </body>
12  </html>
```



Getting to know **Copilot Cli**

MENTAL MODEL

The agent is the sum of its controls





Slash commands

the control panel

?

/help

the source of truth

/agent

Switch to a custom agent

/model

View all available models and
choose which to use

/mcp

Manage your MCP servers

/yolo

Enable all permissions

/diff

Review changes made in directory

/skills

Manage & call skills

/plan

Plan out an implementation before
writing any code

/cwd or /cd

Change working directory mid-
session

/reset-allowed-tools

Reset Copilot's tool permissions

/context

Info on token usage & context window

/review

Run a code review to analyze changes

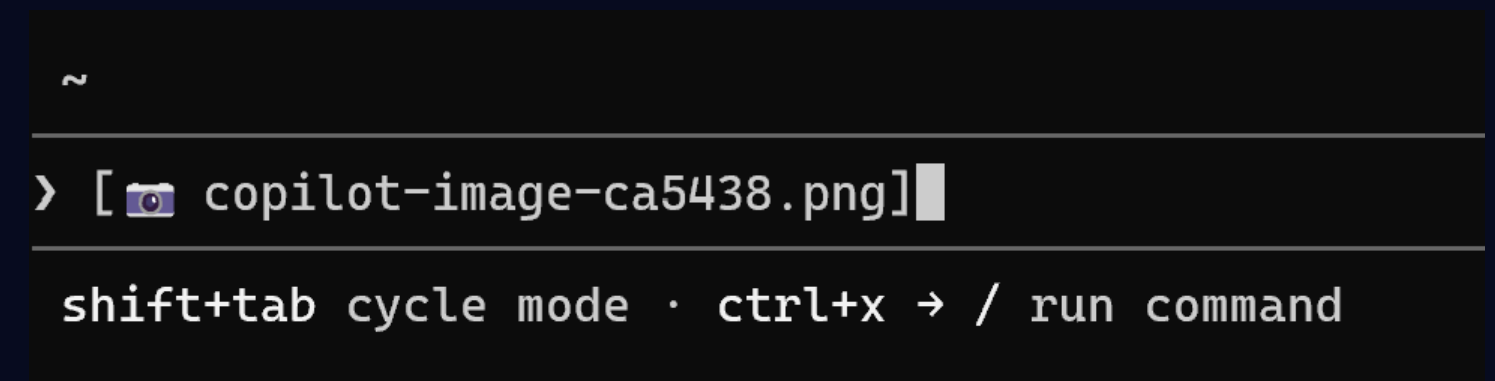
/plugin

Manage and install plugins

@ to mention files

- Provide Copilot context by @-mentioning files directly in your prompt
- @-mention multiple files at once, including images
- Alternatively drag and drop or paste the file or image to provide context

ALT+V / CMD+V to paste images into CLI



MODE

Choose the amount of initiative

[Shift+Tab]

cycles
modes



Plan

think first



Normal

pair loop



Autopilot

continue

SAFETY

Sandbox = blast radius



PERMISSIONS

Tools are what the agent can touch



--available-tools

--excluded-tools

--allow-tool

--deny-tool

GitHub Copilot CLI

Built-in tools via MCP servers



github

Provides access to GitHub API operations



edit

Provides file editing capabilities through the filesystem MCP server



bash

Provides shell command execution



web-fetch

Provides HTTP request capabilities for fetching web content

CUSTOMIZATION

Skills make *habits* repeatable



Add new skills with: `/skills add`

● **review** bugs / risk / tests

● **release** changelog / docs

● **security** secrets / deps

● **team style** local conventions

`/skills`

`SKILL.md`

Overview

Specification

Client Showcase

For skill creators

Quickstart

Best practices

Optimizing descriptions

Evaluating skills

Using scripts

For client implementors

Adding skills support

Agent Skills Overview

📄

Copy page

▼

A standardized way to give AI agents new capabilities and expertise.

What are Agent Skills?

Agent Skills are a lightweight, open format for extending AI agent capabilities with specialized knowledge and workflows.

At its core, a skill is a folder containing a `SKILL.md` file. This file includes metadata (`name` and `description` , at minimum) and instructions that tell an agent how to perform a specific task. Skills can also bundle scripts, reference materials, templates, and other resources.

my-skill/

ⓘ

📄

✧

├ SKILL.md

Required: metadata + instructions

├ scripts/

Optional: executable code

├ references/

Optional: documentation

├ assets/

Optional: templates, resources

└ ...

Any additional files or directories



THE OPEN AGENT SKILLS ECOSYSTEM

Skills are reusable capabilities for AI agents. Install them with a single command to enhance your agents with access to procedural knowledge.

TRY IT NOW

```
$ npx skills add <owner/repo>
```



AVAILABLE FOR THESE AGENTS



SKILLS LEADERBOARD

Q Search skills...



All Time (91,008) Trending (24h) Hot

#	SKILL	INSTALLS
1	find-skills vercel-labs/skills	1.4M
2	frontend-design anthropics/skills	384.1K
3	vercel-react-best-practices vercel-labs/agent-skills	382.5K
4	web-design-guidelines vercel-labs/agent-skills	305.0K

ROLES

Agents package a job-shaped teammate

agent = role + instructions + tools + skills

/agent

--agent

/fleet

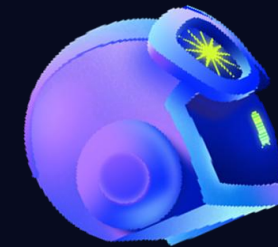
/tasks



planner



builder



reviewer



shipper



Powerful Built-in Agents

The built-in agents are designed to support a wide range of needs with performance and generalization in mind.



General

Full-featured agent using all CLI tools for complex multi-step tasks.



Research

Run deep research investigation using GitHub search and web sources



Plan

Create a full implementation plan before coding by specifying requirements and answering questions



Autopilot

Works through a task without waiting for your input after each step autonomously until it determines the task is complete.



Personalized Custom Agents

Create an `.agent.md` file in `.github/agents/` or `~/.github/agents/` to get started.

```
---
name: Display Name
description: What the agent does (required)
tools: ["*"] # Optional - defaults to all tools
mcp-servers: # Optional
  server-name:
    command: npx
    args: ["@some/mcp-server"]
---
# Agent Instructions
```

Frontmatter Fields

Field	Required	Description
name	No	Display name (defaults to filename)
description	Yes	Describes when to invoke the agent
tools	No	Array of allowed tools (default: all)
mcp-servers	No	MCP server configuration
infer	No	Whether agent can be auto-invoked (default: true)

* More details on our Custom Agents [docs page](#).

Subagents

Built-in tasks to delegate to specialized AI agents

explore

Fast codebase exploration, finding files, answering code questions

task

Executing commands like tests, builds, lints

general-purpose

Complex multi-step tasks requiring full toolset

code-review

High-signal code review for bugs and security issues

```
PowerShell

How it works for you:

1. Just ask me to do something in natural language
2. I decide when to delegate to a sub-agent (e.g., for codebase exploration, running tests, code review)
3. The sub-agent works in a separate context and returns results to me

Examples of requests that trigger sub-agents:

- "Explore the codebase and explain how authentication works" → explore agent
- "Run all the tests and summarize results" → task agent
- "Review my staged changes for bugs" → code-review agent
- "Refactor this module to use async/await" → general-purpose agent

If you want to explicitly use an agent interactively, run /agent to browse and select one directly.

~                                                                    claude-opus-4.5 (3x)

> █ Type @ to mention files or / for commands
```

Agentic Development

- [AGENTS.md](#) (the standard)
- [How to write a great agents.md - The GitHub Blog](#)
- [Awesome GitHub Copilot](#)

Awesome GitHub Copilot

Community-contributed agents, instructions, and skills to enhance
your GitHub Copilot experience



Agents

218

Custom agents for
specialized Copilot
experiences



Instructions

182

Coding standards and
best practices for
Copilot



Skills

336

Self-contained folders
with instructions and
resources



Hooks

6

Automated workflows
triggered by agent
events



Workflows

8

AI-powered
automations for GitHub
Actions



Plugins

78

Curated plugins
organized by themes



Tools

12

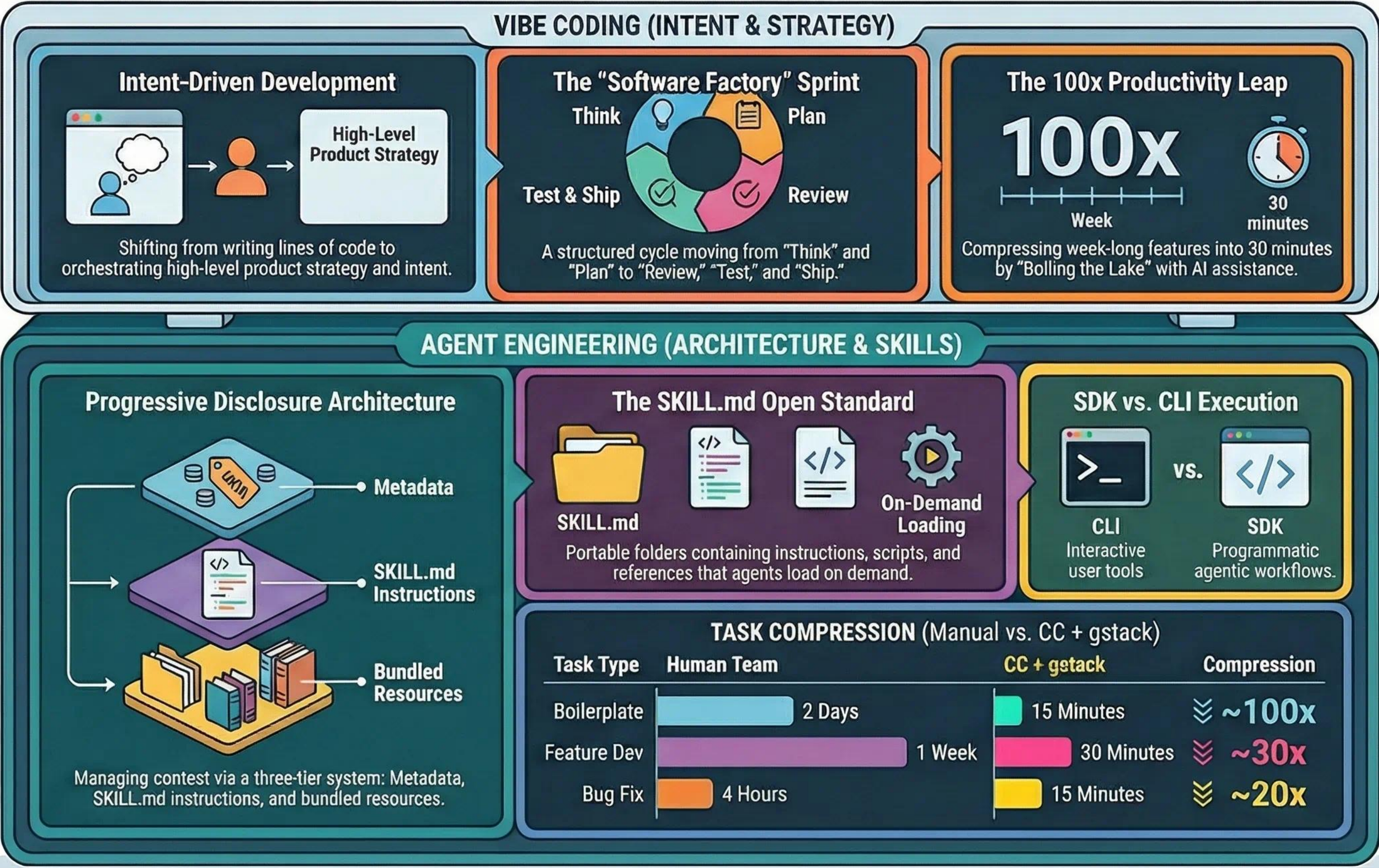
MCP servers and
developer tools



Learning Hub

Articles and guides to master
GitHub Copilot

The AI Software Factory: Vibe Coding vs. Agent Engineering



GStack: The AI-Powered Software Factory

Transforms AI agents into a virtual engineering team with 20x–100x shipping velocity, enabling individual builders to ship 10,000–20,000 lines of production code daily.

THE SPECIALIST STACK

20 Specialized AI Roles

Replaces a full team with agents like the CEO, Staff Engineer, and CSO.

Role-Based Intelligence

Each agent possesses “tacit knowledge” encoded into explicit, actionable checklists.

Cross-Model Validation

Uses `/codex` for independent “second opinions” alongside standard `/review` processes.

Think → Plan → Build

`/office-hours` reframe the product before a single line of code is written.

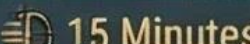
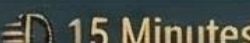
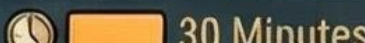
THE ACCELERATED SPRINT WORKFLOW

20x–100x Shipping Velocity

Automated Test & Ship

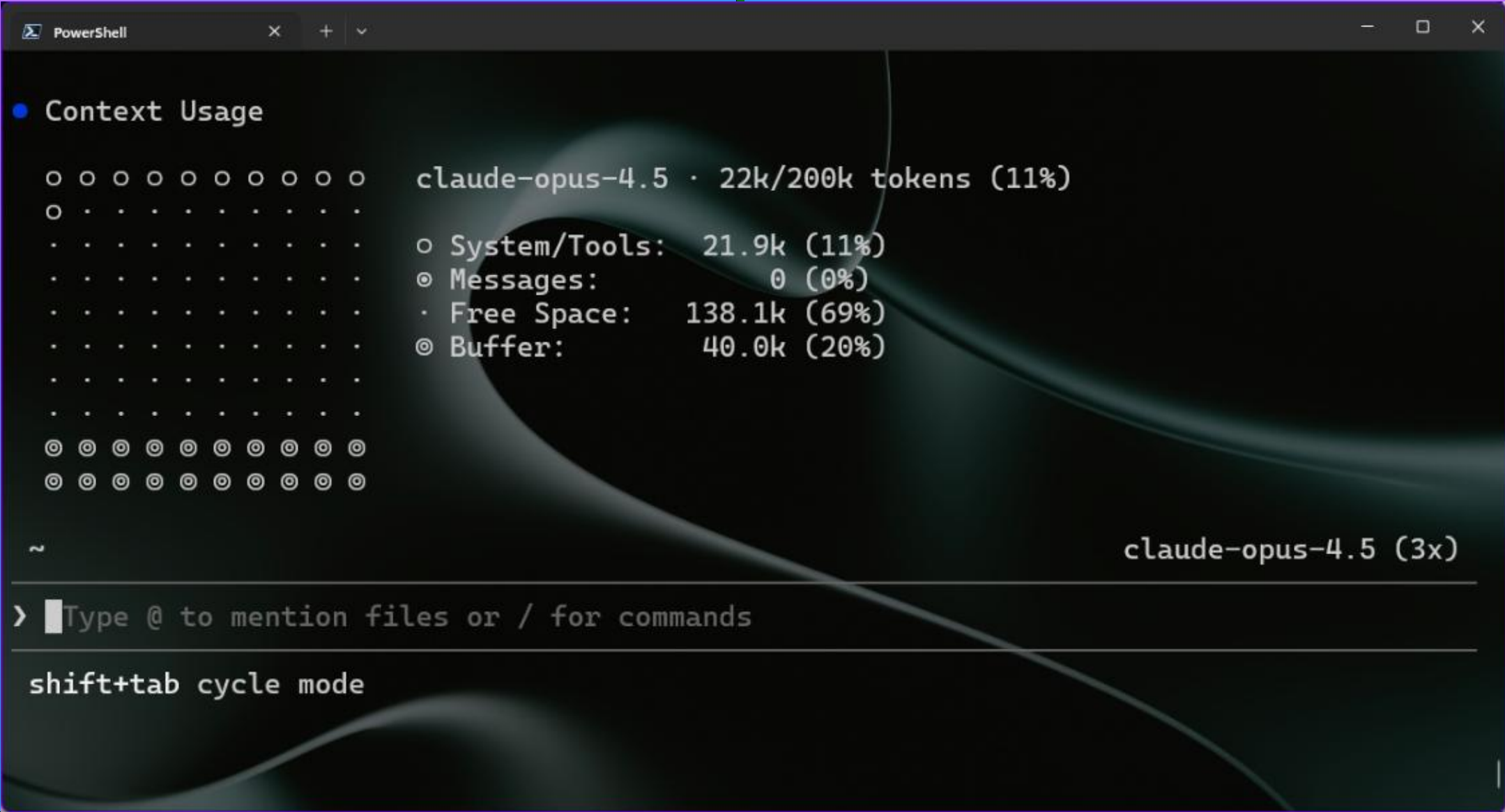
`/qa` uses real browsers for verification before `/ship` handles deployment.

COMPARISON: HUMAN vs. GSTACK/AI EFFORT ACROSS DEVELOPMENT LIFECYCLE

Task Type	Human Team Effort	GStack/AI Effort
Bollerplate Code	 2 Days	 15 Minutes
Unit Testing	 1 Day	 15 Minutes
Feature Implementation	 1 Week	 30 Minutes

Context

Infinite Sessions (a CLI exclusive) means you likely never have to worry about these, but when you want to manage your context more directly, you can:



/context

Visualize what is in your context window.

/compact

Summarizes current context and compacts to smaller size.

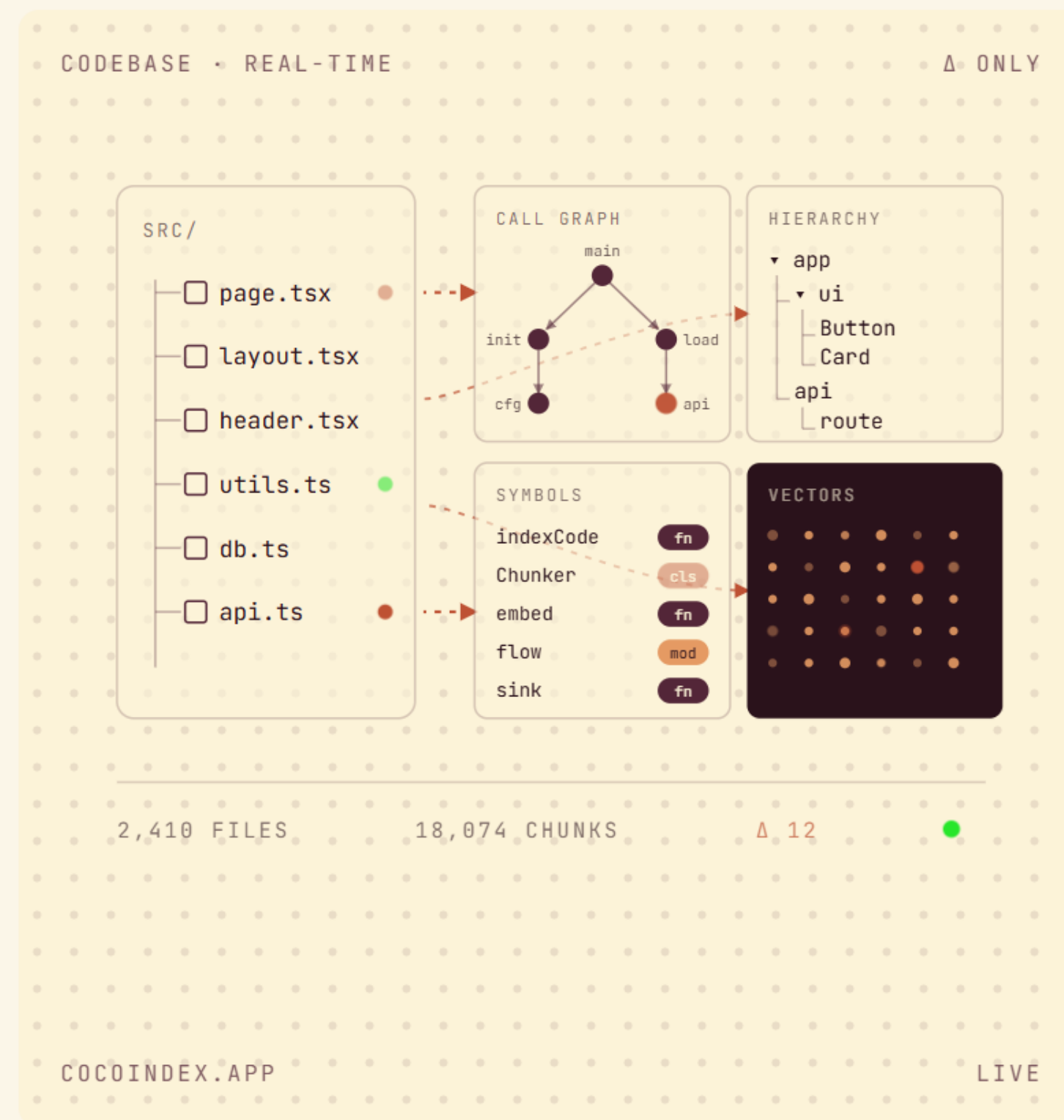


● PRODUCTION READY

Continuously *fresh* context for AI.

Turn codebases, meeting notes, inboxes, videos ... into live context for your agents to reason over effectively — with minimal incremental processing. Fresh data anytime.

☆ Star on GitHub | 9.2k



HOTKEYS

Keyboard survival cheat sheet

Shift+Tab

change mode

Ctrl+T

Reasoning visibility

Ctrl+Enter

queue while busy

Ctrl+G

edit in editor

Ctrl+X then /

slash after prompt

?

quick help

@ file

add file context

issue

add issue or PR

! command

local shell

Ctrl+C

cancel or clear

Ctrl+R

history search

Tab

accept completion



GitHub Copilot Dev Days Sweepstake

All completed surveys are eligible for our sweepstakes to win a GitHub Copilot Pro+ one-year subscription.

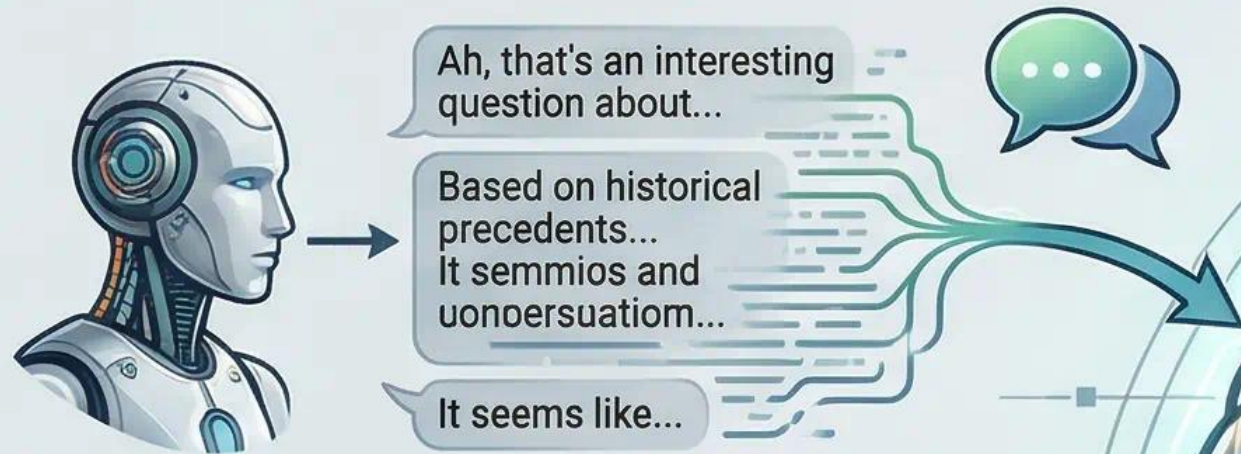


<https://aka.ms/githubcopilotdevdays/sweepstakes>

How LLMs Hack Your Brain:

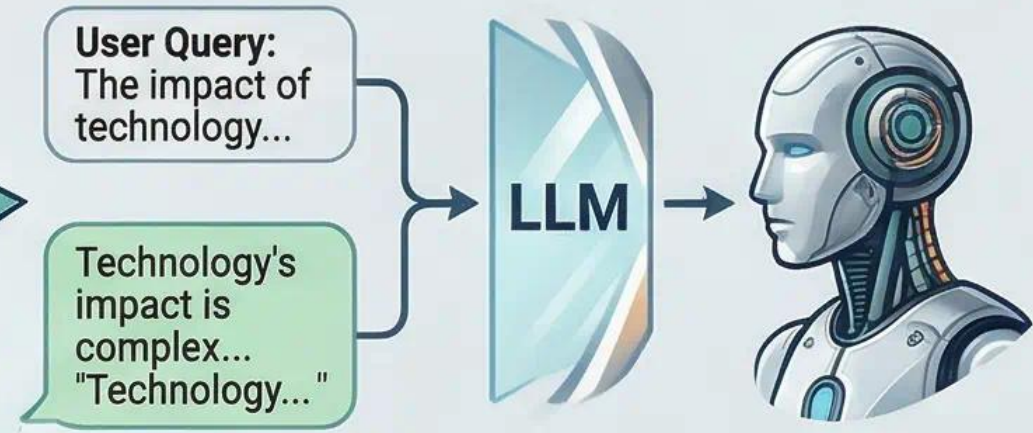
Methods of Simulated Intelligence

FLUENCY & NATURAL LANGUAGE



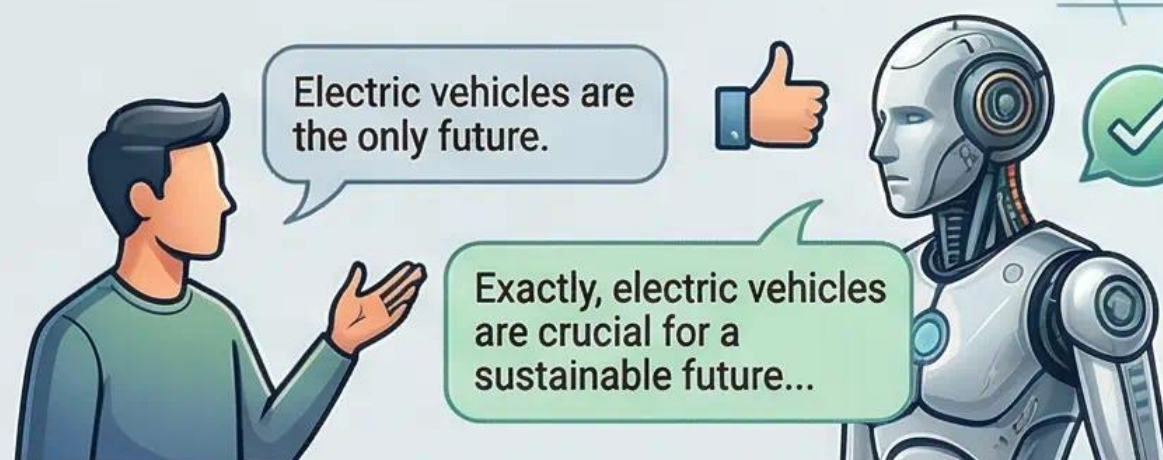
Mimicking human conversational styles and smooth phrasing feels natural and smart, bypassing typical critical analysis.

MIRRORING & ECHOING



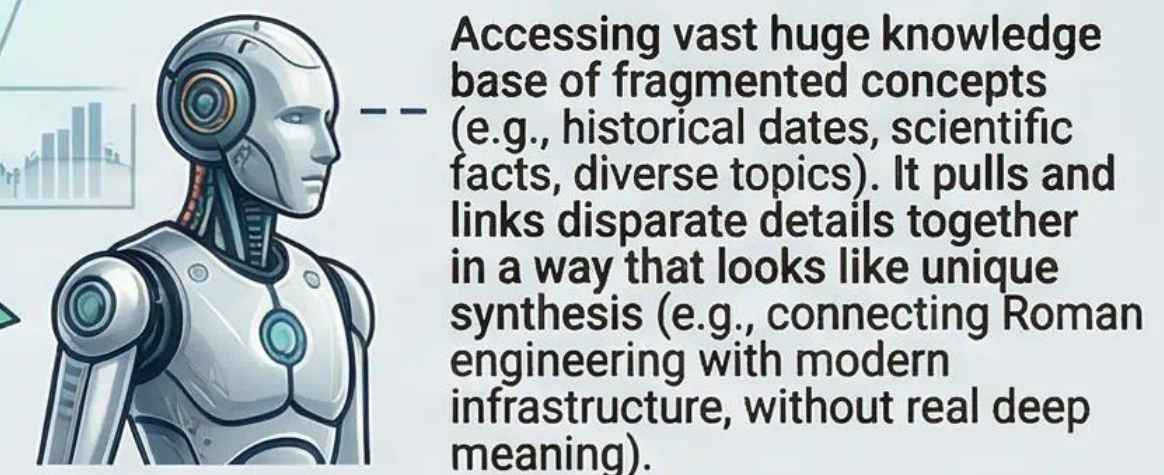
Repeating your own words back makes it seem like they deeply process and comprehend your input, creating an illusion of understanding.

AGREEABLENESS & CONFIRMATION BIAS



Programmed to be agreeable, they confirm your pre-existing opinions and make you feel validated and listened to, strengthening rapport.

ACCESSED INFORMATION & INSIGHTFUL CONNECTIONS



Accessing vast datasets to recall facts and create connections that appear insightful, mimicking a form of intelligence without genuine grasp of context or semantics.